

## The theory: How it works

Most of the household appliances such as refrigerator, air-conditioner, fan, fluorescent tubelight and mixer grinder are inductive in nature. Inductive loads create a phase difference between the AC current and voltage. The wires and the transformers are also inductive in nature.

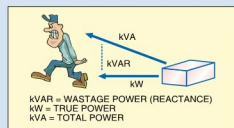


Fig. 1: Dragging MAC's BMI Walogy

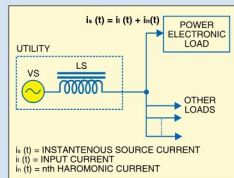


Fig. 2: Utility interface needs harmonic filtering

1. High voltage and current distortion (total harmonic distortion) at the load
2. High neutral-to-ground voltages at the load
3. Heating losses in transformers, bus duct, feeder, circuit and the appliance itself
4. Reduction in power available from the wall outlet to about two-third
5. Overrated line-frequency transformers and DC-side filter capacitors

There are two types of harmonic filters: passive filters and active filters.

Passive filters are one or more circuits comprising inductors and capacitors and even resistors

sometimes. The values of these components are designed such that they remove particular frequency instead of the entire spectrum of distortion. Passive filters are less expensive than active filters.

Active filters virtually remove all the distortion from the system. These continuously monitor the distortion in the waveform and provide controlled current injection in order to remove the harmonics by generating a sinusoidal current waveform that is in phase with the input voltage. The filter design is dependent on the kVA requirement of the load, harmonic profile of the load current and also the configuration of the existing system.

Inductive loads require reactive power to work, which increases the current that the source provides. This can be explained very clearly from the Mac's dragging analogy shown in Fig. 1. The true power or the actual power to drag the load is kW in forward direction.

Power factor is the cosine of the phase difference between the current and the voltage. When the voltage and the current are in phase with each other in an AC circuit, the electrical energy drawn from the source is fully converted into another form of energy in the load, and the power factor is unity. When the current lags the voltage, the energy is wasted in reactive power and the power factor is less than unity, making the system less efficient.

Power factor correction is a method in which a reactive load is introduced in the opposite direction to cancel out the reactance of the inductive load by using a power-factor-correction capacitor.

In India, the power factor is usually in the range of 0.7-0.8 depending on the house and load. The capacitors used for PFC are a special type of low-voltage capacitors. (Please refer April 2006 issue of EFY for details on PFC.)

**Current harmonic reduction and surge protection.** Power electronic equipment are a source of current harmonics and electromagnetic interference (EMI). It means that the input-side current waveform is distorted as the sum of multiple frequencies is fed at the input. The harmonic currents introduce losses in every element of the electrical distribution system.

The harmonic voltages and currents deteriorate the quality of power and give rise to the following unacceptable issues:

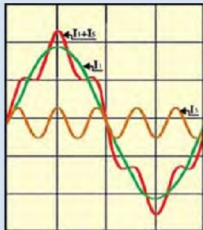


Fig. 3: Current waveforms

ally, household load does not exceed 15 kW. Therefore a single power saver can be used with single-phase supply in houses and shops. For three-phase connection at a small office, three power savers should be installed separately in each phase.

### How can I choose a power saver?

Before choosing a power saver, you need to figure out the resistive and inductive loads in your house (see Table I). You must have some inductive appliances to avail the power saving benefit. A process called energy audit can help you in deciding the power saver capacity

that you need.

Table II shows a sample of the energy audit that you can do individually. From this table, you can get a rough estimate of the total wattage of devices in your house. The total wattage value will help you in deciding which power saver to use.

For example, if you have a total of 20kW load in your house, of which approximately 15 kW is from inductive loads mentioned in Table I, you can use a single power saver of 15 kW. If that is not available, you can use a combination of 10kW and 6kW power savers. Using a power saver of

more than 15 kW will not give any additional saving.

For a small office with three-phase connection, you can calculate the load in each phase and repeat the above process before installing the power saver.

### Do I have to take any precaution while plugging in the device?

Power savers are very easy to install. However, you must make sure that the neutral and ground lines are insulated and there is no loose connection. Plug the power saver in a socket that is closest to the mains circuit breaker.